



LED As LIGHT SENSOR

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Usually light-dependent resistors (LDRs) and photodiodes are used as light sensors to activate or deactivate circuits based on the presence or absence of light. But a common transparent LED can also be used in the place of an LDR or a photodiode. Here is a simple circuit that uses a transparent red LED as the light sensor that switches on a white LED when the red LED is in dark.

Ordinary transparent LEDs function as light emitters and light detectors. These detect a narrow band of wavelengths of light and their p-n junction forward biases to generate around 1V voltage with a minute current of 0.030mA when exposed to bright sunlight. So a transparent LED can also be used as a light sensor like a photodiode.

LEDs detect a much narrower band of light having peak sensitivity at a wavelength slightly lower than the wavelength these emit. For

example, a transparent red LED emits around 660nm light and responds better to orange light at 610nm.

Circuit and working

Circuit diagram of an LED as a light sensor is shown in Fig. 1. It is built around op-amp CA3140 (IC1), transistors BC547 (T1) and BC337 (T2).

The circuit uses high-gain CA3140 to amplify the feeble current generated by the LED in the presence of sunlight. CA3140 is a high-performance 4.5MHz BIMOS operational amplifier having MOSFET inputs and bipolar output.

Gate-protected MOSFET transistors in the input circuit provide very high input impedance, typically around 1.5-tera-ohm. These require very low input current (as low as 10-pico-ampere) to change their output status. So even without external bias, IC1 performs well. Here, transparent LED1 and its low current generated will bias IC1 when exposed to light.

In the circuit, transparent red LED1 is connected in reverse-bias mode with biasing resistor R1. IC1 is wired as a simple current-to-voltage converter. It is a trans-impedance amplifier with resistor R2 to set the current feedback. So when sunlight falls on LED1, it generates

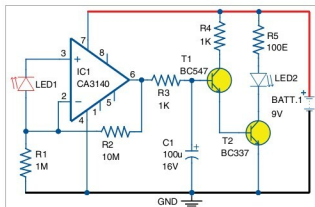


Fig. 1: Circuit diagram of an LED as a light sensor

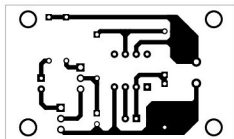


Fig. 2: Actual-size PCB layout of the LED as a light sensor circuit

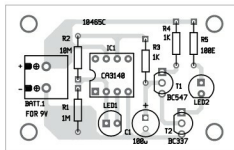


Fig. 3: Component layout of the PCB

PARTS LIST

Semiconductors:

- IC1 - CA3140 op-amp
- T1 - BC547 npn transistor
- T2 - BC337 npn transistor
- LED1 - 5mm transparent red LED
- LED2 - 5mm white LED

Resistors (all 1/4-watt, $\pm 5\%$ carbon):

- R1 - 1-mega-ohm
- R2 - 10-mega-ohm
- R3, R4 - 1-kilo-ohm
- R5 - 100-ohm

Capacitors:

- C1 - 100 μ F, 16V electrolytic

Miscellaneous:

- CON1 - 2-pin terminal connector
- 9V battery

a minute reverse current that keeps the voltage at the inverting input of IC1 slightly higher. This keeps output of IC1 low. As a result, T1 and T2 remain off to turn off white LED2.

When LED1 is in dark, reverse current ceases and suddenly IC1 changes its output to high state. This turns on T1, followed by T2. The white LED2 connected to the collector of T2 turns on.

Construction and testing

An actual-size, single-side PCB layout for the circuit is shown in Fig. 2 and its component layout in Fig. 3. Assemble the circuit on the PCB and enclose in a suitable box.

Light sensor LED1 should be a high, bright, transparent 5mm red LED. Transparent green and white LEDs also work well but red LEDs show more sensitivity towards ambient light. Light from white LED2 should not fall on red LED1 sensor. In place of the white LED, a relay can be used to switch on AC loads, so that the circuit functions as an automatic nightlight.

The circuit works off a 9V battery power supply. **EFY**



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